Cox Proportional Hazards Regression

Cox Proportion Hazards Regression

Multivariable

Examine multiple predictors simultaneously

Estimate Effect Size

Hazard Ratio = Hazard in one group / Hazard in reference group

Hazard: Rate at which events occur at a given time among those who have not yet experienced the event

Demo

Cox Proportional Hazards Regression

Knowledge Check

True or False: According to this model, participants with a depression diagnosis have a higher hazard of death than those without depression, after adjusting for the other predictors.

```
Call:
coxph(formula = Surv(mos_to_death, died) ~ female + age_2014 +
    bp_dx + depress_dx, data = hrs_df)

               coef exp(coef)  se(coef)      z        p
female      0.393772  1.482562  0.086492  4.553  5.3e-06
age_2014    0.290459  1.337042  0.009521 30.508  < 2e-16
bp_dx      -0.058589  0.943094  0.089292 -0.656 0.511726
depress_dx  0.375581  1.455837  0.103778  3.619 0.000296

Likelihood ratio test=1063  on 4 df, p=< 2.2e-16
n= 2355, number of events= 660
   (62 observations deleted due to missingness)
```

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Proportional Hazards Assumption

The hazard ratio between groups is assumed to remain constant over time.

Test

H_0: The proportional hazards assumption holds.
H_a: The proportional hazards assumption is violated.

Remember: crossing Kaplan-Meier curves may suggest a violation, but formal testing is needed.

Demo

Check the Proportional Hazards Assumption: *Are hazard ratios constant over time?*

Knowledge Check

True or False: Based on the global test, there is no statistically significant evidence that the proportional hazards assumption is violated overall.

```
              chisq df     p
female      0.00396  1 0.950
age_2014    4.33717  1 0.037
bp_dx       0.45907  1 0.498
depress_dx  0.00075  1 0.978
GLOBAL      5.74540  4 0.219
```

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